Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY

9700/01

Paper 1 Multiple Choice – The Microscope in Cell Studies

MARK SCHEME

Maximum Mark: 29

Published

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Question	Answer	Marks
1	**B** ; *Statement 2 is correct because resolution of a light microscope is fundamentally limited by the wavelength of visible light (~400-700 nm, giving min resolution ~200 nm). Magnification is limited by lenses, not wavelength. Statement 3 is incorrect because total magnification is the product (eyepiece × objective), not sum.*	1
2	**C** ; *An electron microscope has both higher magnification (> ×500 000 vs ×1 500) and higher resolution (0.5 nm vs 200 nm) due to the much shorter wavelength of an electron beam compared to visible light.*	1
3	**C** ; *Option C correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
4	**D** ; *Methylene blue is an animal cell stain; onion epidermal cells are traditionally stained using iodine solution (in KI) to visualize cell walls and nuclei.*	1
5	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
6	**C** ; *Ribosomes have an average diameter of ~20-30 nm (70S/80S), which is well below the 200 nm resolution limit of a standard light microscope. Mitochondria (~1 μm), lysosomes (~0.4 μm) and chloroplasts (~5 μm) are all > 200 nm and can be resolved.*	1
7	**C** ; *Option C correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
8	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
9	**C** ; *To convert from micrometres (μm) to nanometres (nm), multiply by 1 000: 0.4 μm × 1 000 = 400 nm.*	1
10	**C** ; *Resolution is defined as the shortest (minimum) distance at which two separate objects/points can still be distinguished as distinct entities.*	1
1	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
2	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
3	**C** ; *Option C correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1

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Question	Answer	Marks
4	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
5	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
6	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
7	**C** ; *Option C correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
8	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
9	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
10	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
1	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
2	**B** ; *Option B correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
3	**C** ; *Option C correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
4	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
5	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1

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Question	Answer	Marks
6	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
7	**C** ; *Option C correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
8	**D** ; *Option D correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1
9	**A** ; *Option A correctly fulfills the Cambridge 9700 learning criteria for The Microscope in Cell Studies. In accordance with syllabus standards, the distractors represent common misconceptions or inverted relationships.*	1

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